

Table 1

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(dplyr)
library(broom)
library(tinytex)
library(gtsummary)
library(kableExtra)

##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows

clean_data <- read.csv("data/clean_state_data.csv")

clean_data$phf_per_capita[is.na(clean_data$phf_per_capita) & clean_data$state == "Georgia"] <- 108
clean_data$phf_per_capita[is.na(clean_data$phf_per_capita) & clean_data$state == "Mississippi"] <- 5678

clean_data_table1 <- clean_data %>%
  rename("% Insured" = pct_insuredE,
         "% with Bachelor's Degree or higher" = pct_bachelors_plusE,
         "% Living Under Poverty Line" = pct_povertyE,
         "% Living in Urban Area" = pct_urban,
         "Public Health Funding per Capita" = phf_per_capita) %>%
  mutate(exposure = case_when(
    exposure_binary == 0 ~ "Non-Permissive Exemptions",
    TRUE ~ "Permissive Exemptions"
  ))

clean_data_table1 %>%
  tbl_summary(include = c(`% Insured`, `% with Bachelor's Degree or higher`, `% Living Under Poverty Line`,
    by = exposure, type = all_dichotomous() ~ "categorical", missing_text = "NA") %>%
  bold_labels() %>%
  modify_caption("Demographic Characteristics of 50 US States + Washington DC, 2024") %>%
  as_kable_extra(booktabs = TRUE, longtable = TRUE, linesep = "") %>%
  kableExtra::kable_styling(position = "center", latex_options = "hold_position", font_size = 9) %>%
  column_spec(2:3, width = "3.5cm")

## Warning in stri_detect_regex(string, pattern, negate = negate, opts_regex =
## opts(pattern)): argument is not an atomic vector; coercing
```

Table 1: Demographic Characteristics of 50 US States + Washington DC, 2024

Characteristic	Non-Permissive Exemptions N = 34	Permissive Exemptions N = 17
% Insured	92.70 (91.00, 94.50)	93.30 (90.80, 94.20)
% with Bachelor's Degree or higher	36.2 (32.4, 41.2)	34.6 (32.4, 37.8)
% Living Under Poverty Line	11.45 (10.10, 12.60)	11.70 (10.30, 13.40)
% Living in Urban Area	73 (62, 87)	73 (67, 83)
Public Health Funding per Capita	52 (37, 94)	45 (32, 71)
NA	2	1

¹ Median (Q1, Q3)

Outcome Map

```
library(sf)

## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE

library(ggplot2)
library(dplyr)
library(maps)

##
## Attaching package: 'maps'

## The following object is masked from 'package:purrr':
##
## map

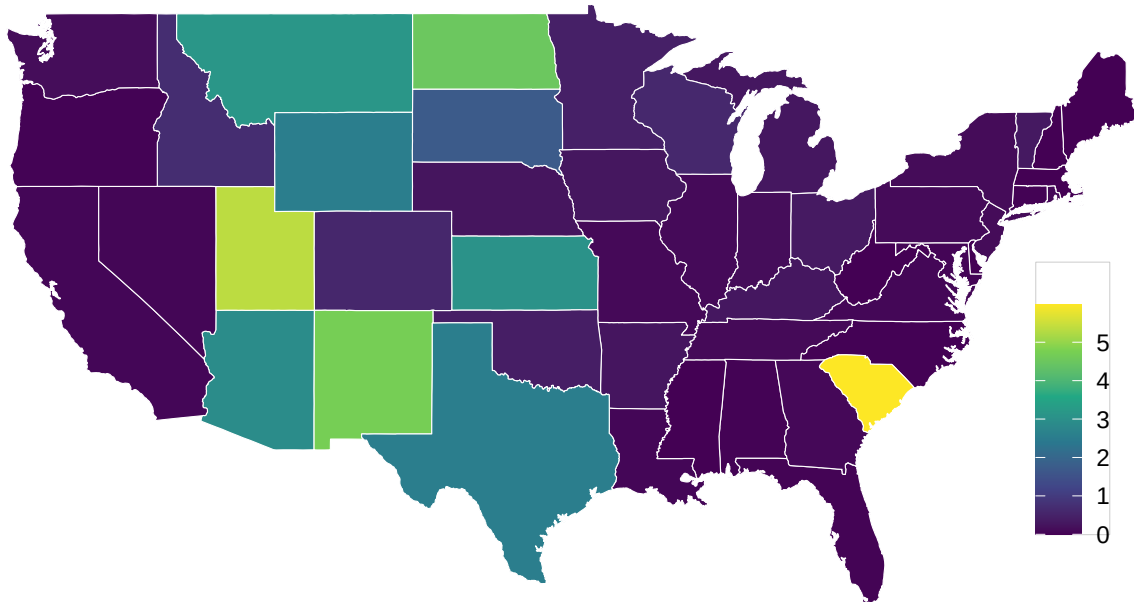
states <- st_as_sf(map("state", plot = FALSE, fill = TRUE))

# The map package uses lowercase state names
states$state <- tools::toTitleCase(states$ID)

# Join your data to map geometries
map_df <- left_join(states, clean_data, by = "state")

# Plot
ggplot(map_df) +
  geom_sf(aes(fill = cases_per_100k, color = "white")) +
  scale_fill_viridis_c(option = "D") +
  theme_void() +
  labs(fill = "") +
  theme(
    legend.position = c(0.87, 0.15), # x (left), y (down from top)
    legend.justification = c(0, 0),
    legend.background = element_rect(fill = "white", color = "gray80"),
    plot.title = element_text(hjust = 0.2),
    legend.key = element_rect(fill = NA, colour = NA)
  ) +
  ggtitle("2025 Measles Case Rate per 100k in the US")
```

2025 Measles Case Rate per 100k in the US



```
#binning might be helpful
data_bin <- clean_data %>%
  mutate(bin_case = case_when(
    measles_cases == 0 ~ "0",
    measles_cases >= 1 & measles_cases < 10 ~ "1-9",
    measles_cases >= 10 & measles_cases < 40 ~ "10-49",
    measles_cases >= 50 & measles_cases < 100 ~ "50-99",
    measles_cases >= 100 & measles_cases < 500 ~ "100-499",
    TRUE ~ "500+"
  ))

map_df_bin <- left_join(states, data_bin, by = "state")

# Plot
ggplot(map_df_bin) +
  geom_sf(aes(fill = bin_case), color = "white") +
  scale_fill_brewer(palette = "Blues") +
  theme_void() +
  labs(fill = "") +
  theme(
    legend.position = c(0.87, 0.15), # x (left), y (down from top)
    legend.justification = c(0, 0),
    legend.background = element_rect(fill = "white", color = "gray80"),
    plot.title = element_text(hjust = 0.2),
    legend.key = element_rect(fill = NA, colour = NA)
  ) +
  ggtitle("2025 Measles Case Counts in the US")
```

2025 Measles Case Counts in the US

